

Review

Artificial intelligence in multidisciplinary tumor boards enhancing decision making and clinical outcomes in oncology

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SUMMARY

Multidisciplinary tumor boards (MDTs) coordinate complex oncology decisions across imaging, pathology, genomics, and patient factors. Here we synthesize how artificial intelligence (AI)—including machine learning, natural language processing, deep learning, and large language models—supports MDT preparation and deliberation. Across cancers, reported agreement between AI recommendations and MDT decisions commonly ranges from 70% to 90%, and task-focused tools achieve high diagnostic performance in screening and staging. Benefits include faster information synthesis, more consistent guideline alignment, and clearer documentation of options, while human review remains central. Key limitations—data bias, uneven generalizability, privacy and governance concerns, and limited prospective validation—temper adoption. We outline implementation priorities: prospective multicenter evaluation, integration with electronic records, clinician training, and transparent oversight. Overall, AI can augment MDT decision-making and help personalize care and workflow efficiency when deployed with rigorous evaluation and safeguards.

INTRODUCTION

Cancer remains a leading cause of morbidity and mortality, with tens of millions of new cases annually.¹ Common solid tumors demand nuanced decisions across diagnosis, staging, and therapy selection. Multidisciplinary tumor boards (MDTs), which convene oncologists, surgeons, radiologists, and pathologists, are central to contemporary cancer care.² By pooling expertise, MDTs improve diagnostic accuracy, raise guideline adherence, and tailor treatment plans that enhance outcomes.³

However, MDTs face practical and cognitive constraints. Coordinating busy specialists is resource-intensive and can delay treatment.⁴ High case volumes risk decision fatigue and oversight, while ambiguity in radiology or pathology introduces inter-expert variability.^{5,6} Evidence evolves rapidly—new biomarkers, trial readouts, indications, and toxicity data—complicating real-time retrieval during meetings.⁷ Data silos and inconsistent documentation further hinder synthesis of the full clinical picture.⁸

Artificial intelligence (AI) offers tools to augment, not replace, MDT deliberation. Across machine learning (ML), natural language processing (NLP), deep learning (DL), and large language models (LLMs), AI systems analyze heterogeneous inputs to surface patterns, extract salient facts, and support guideline-concordant reasoning.⁹ In oncology, ML automates staging and risk stratification¹⁰; NLP composes concise summaries

from unstructured notes and reports¹¹; DL excels at image-based detection and characterization¹²; and LLMs generate transparent, evidence-linked recommendations and patient-facing documents¹³ (Figure 1).

Early evaluations suggest AI aligns closely with expert MDT decisions while reducing cognitive load.¹⁴ Reported AI-MDT concordance frequently falls in the 70–90% range across several cancers.¹⁵ NLP pipelines identifying high-risk pancreatic lesions have achieved near-perfect sensitivity with excellent specificity.¹⁶ Cognitive decision-support platforms show moderate-to-high agreement with MDT treatment choices in colorectal and lung cancer, and iterative prompting of LLMs improves concordance for complex gastrointestinal cases.¹⁷ Standardizing inputs reduces interobserver variability and focuses discussion on truly contentious judgments.¹⁸

Limitations persist: uneven performance in underrepresented subgroups or atypical cases (data bias, limited generalizability), occasional non-recommendation with incomplete or rare presentations, and opaque behavior undermining trust¹⁸; prospective real-world studies are scarce, requiring robust privacy safeguards, human oversight, and clear accountability.¹⁶

This review synthesizes evidence across four domains—diagnostic and staging support, treatment recommendation and personalization, prognosis and risk assessment, and cancer-specific applications. We compare modalities, summarize representative performance, and examine concordance with human experts. We also analyze technical, ethical, regulatory, and



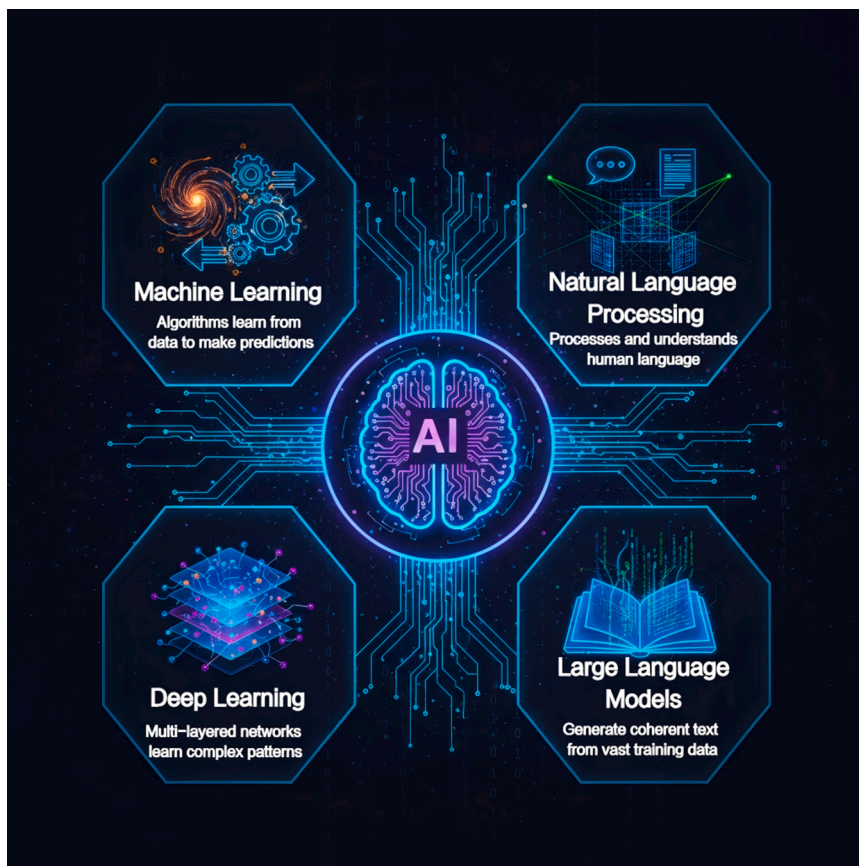


Figure 1. Core Concepts of AI Technologies in Oncology

This conceptual diagram illustrates the four principal paradigms of AI applied in oncology and multidisciplinary tumor board decision support—machine learning, natural language processing, deep learning, and large language models.

ating characteristic curve [AUC] 0.94 for chemotherapy decisions, 0.977 for targeted therapy) by learning from past MDT outcomes.²³

Natural language processing

NLP techniques enable extraction of structured information from unstructured text.³⁰ Early rule-based NLP systems and newer transformer-based models can interpret medical language to support MDT preparations.³¹ Key applications include auto-aggregating patient history from electronic health records (EHRs), flagging high-risk findings for case discussion, and ensuring semantic consistency using medical ontologies (like Systematized Nomenclature of Medicine – Clinical Terms [SNOMED CT]).³² For instance, an NLP tool scanning free-text radiology reports identified patients with pancreatic cysts who might benefit from MDT review, with 99.9% sensitivity and 98.8% specificity in detecting those at risk.¹⁶ By synthesizing such

data, NLP helps ensure that no relevant detail is overlooked during the tumor board meeting.

Deep learning

DL uses multi-layered neural networks capable of learning complex patterns from high-dimensional data such as medical images, genomic profiles, or multi-omics data.³³ Convolutional neural networks, for example, excel at image analysis and have been applied to pathology slides and radiology scans to identify tumor features.³⁴ In the MDT context, DL models contribute to prognosis prediction and risk stratification by automatically quantifying disease characteristics.³⁵ As an example, a DL model analyzing magnetic resonance imaging (MRI) scans across multiple centers was able to detect prostate cancer and stratify tumor stage with AUCs around 0.91–0.95.³⁶ Similarly, AI-driven prognostic models for hepatocellular carcinoma have demonstrated accuracy exceeding 80% in predicting survival outcomes, assisting MDTs in prioritizing high-risk cases for aggressive management.³⁷ These results indicate that DL can provide reliable risk estimates and diagnostic classifications to support expert decision-making.

Large language models

LLMs like GPT-4 are advanced AI systems trained on massive text corpora to generate human-like text and infer context.³⁸ In oncology, specialized LLMs and related expert systems can

human-factor challenges and outline next steps. Our goal is a concise, practice-oriented map of where AI can most credibly strengthen MDT decision-making and what evidence and safeguards are needed for broader, equitable adoption in precision oncology. A cross-study summary of datasets, cancer types, validation levels, and performance metrics is provided in Table 1; Figure 2.

OVERVIEW OF AI TECHNOLOGIES IN ONCOLOGY

Machine learning

ML involves algorithms that improve their predictive performance through experience.²⁵ This category includes classical statistical learners (e.g., support vector machines, random forests) as well as ensemble methods.²⁶ In MDT support, ML techniques are used for tasks like tumor classification, outcome prediction, and treatment recommendation.²⁷ ML models can incorporate diverse clinical and non-clinical variables—for example, patient comorbidities or preferences—to enhance decision reliability.²⁸ In one study, an ML model automatically staged lung cancers by applying symbolic rules to pathology reports, achieving 72% accuracy for T-stage, 78% for N-stage, and 94% for M-stage.²⁹ Another ML application predicted oncologists' adjuvant therapy choices in breast cancer with high fidelity (area under the receiver oper-

Table 1. Study summary: dataset size, cancer type, validation level, and performance metrics

Study/context	Cancer type	AI modality	Dataset size (N)	Validation level*	Primary metric	Reported performance	Reference
WFO concordance (Korea)	Colon	Cognitive CDSS	656	Retrospective cohort	AI-MDT concordance	48.9% strict; 88.4% broad	Assessing Concordance With Watson for Oncology, a Cognitive Computing Decision Support System for Colon Cancer Treatment in Korea. ¹⁹
WFO feasibility (China)	Lung	Cognitive CDSS	149 supported (18.1% unsupported overall)	Feasibility (single-center)	AI-MDT concordance	65.8% overall; 100% in stage IV/extensive SCLC	Using Artificial Intelligence (Watson for Oncology) for Treatment Recommendations Amongst Chinese Patients with Lung Cancer: Feasibility Study. ²⁰
LLM vs. MDT	Colorectal (mixed)	LLM (ChatGPT/ Claude/Llama)	95	Case-based evaluation	AI-MDT concordance	Up to 84%	Large Language Models as Decision-Making Tools in Oncology: Comparing Artificial Intelligence Suggestions and Expert Recommendations. ²¹
LLMs for GC MDT	Gastric	LLM (GPT-4o vs. Gemini)	NR	Simulation	AI-MDT concordance	~54.5% → ~84%	ChatGPT-4o outperforms Gemini Advanced in assisting multidisciplinary decision-making for advanced gastric cancer. ²²
Predicting MDT adjuvant decisions	Breast	ML classifiers	1,065 cases	Retrospective (10-fold CV)	AUC (chemo/ endocrine/targeted)	0.940/0.899/0.977	Computational prediction of multidisciplinary team decision-making for adjuvant breast cancer drug therapies: a machine learning approach. ²³
EHR triage of pancreatic cysts	Pancreas	NLP	566,233 reports	Large-scale retrospective screen	Sensitivity/Specificity	99.9%/98.8%	Automated pancreatic cyst screening using natural language processing: a new tool in the early detection of pancreatic cancer. ¹⁶
HCC CDSS (therapy + prognosis)	Liver (HCC)	AI-CDSS/DL	5,642 patients (single system; internal/external splits)	Real-world retrospective development/ validation	Top-1 treatment recommendation accuracy; MDT match; C-index(OS/recurrence)	Top-1 82.36%; MDT match 95.10%; C-index (OS) = 0.735	Research of artificial intelligence-based clinical decision support system for primary hepatocellular carcinoma. ²⁴
Frontier LLM triage	Multi-condition	LLM (GPT-4o/ Claude-3.5/ Gemini-1.5)	144 vignettes	Benchmark comparison	Triage accuracy	92.4%	The Triage and Diagnostic Accuracy of Frontier Large Language Models: Updated Comparison to Physician Performance. ²⁴

AI, artificial intelligence; MDT, multidisciplinary tumor board; CDSS, clinical decision support system; LLM, large language model; ML, machine learning; NLP, natural language processing; DL, deep learning; EHR, electronic health record; HCC, hepatocellular carcinoma; AUC, area under the receiver operating characteristic curve; C-index, concordance index (time-to-event discrimination); OS, overall survival; N, sample size; NR, not reported; CV (10-fold), tenfold cross-validation; Top-1 accuracy, proportion where the AI's highest-ranked option matches the reference (e.g., MDT choice). Validation level legend: Retrospective cohort, analysis on historical cases; Feasibility (single-center), prospective/implementation-oriented pilot at one site; Simulation, offline decision emulation without patient-facing deployment; Case-based evaluation, expert panel comparison on curated cases; Benchmark, standardized vignette/task comparison across methods.

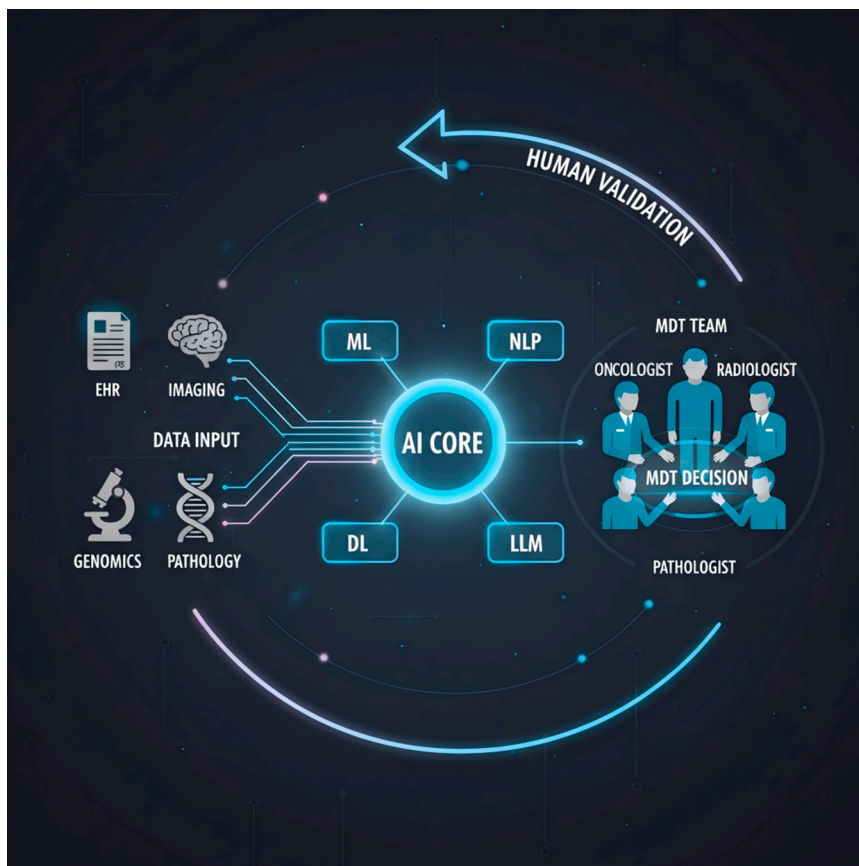


Figure 2. Schematic representation of an AI-integrated multidisciplinary tumor board (MDT) workflow in oncology

The figure illustrates how heterogeneous clinical data—including electronic health records (EHR), medical imaging, pathology reports, and genomic information—are processed through an AI core composed of machine learning (ML), natural language processing (NLP), deep learning (DL), and large language model (LLM) modules. The AI core synthesizes multimodal inputs to generate diagnostic, prognostic, and therapeutic insights that support MDT deliberations among oncologists, radiologists, and pathologists. Final treatment recommendations are reached through human validation, ensuring that AI-assisted outputs are critically reviewed, ethically grounded, and clinically applicable. The circular feedback loop emphasizes continuous learning between AI systems and clinical experts to enhance precision and transparency in oncology decision-making.

SNOMED CT expressions with rule-based logic achieved TNM classification accuracies of ~72% (T), 78% (N), and 94% (M) against expert reference labels in a cohort of 718 patients.²⁹ By programmatically extracting tumor size, nodal involvement, and metastatic status, such tools provide rapid, consistent staging summaries to support MDT review.⁴⁴

NLP-driven prescreening further ensures relevant cases reach the board.

emulate aspects of clinical reasoning. They ingest case details and medical knowledge to suggest treatment options or even draft clinical documents.³⁹ LLMs can categorize recommendations and explain the supporting evidence, improving transparency.⁴⁰ These models have shown moderate-to-high concordance with MDT decisions. For example, Watson for Oncology's (WFO's) treatment suggestions concurred with lung cancer MDT plans about 65–92% of the time (with higher alignment in advanced-stage cases).⁴¹ Meanwhile, a research version of ChatGPT was able to mirror 84% of multidisciplinary decisions in gastrointestinal cancers under certain conditions.⁴² Although LLMs are not yet perfectly consistent, their ability to process clinical guidelines and patient specifics at scale offers a valuable “second opinion” in MDT discussions. They also have been used to generate patient education materials—one study reported an AI model producing consent forms deemed 100% actionable and at an appropriate reading level for patients.⁴³

APPLICATIONS OF AI IN TUMOR MDT DECISION-MAKING

Diagnostic and staging support

Accurate diagnosis and staging underpin MDT decision-making. AI now automates and standardizes key elements of this workflow. In lung cancer, a system leveraging post-coordinated

An algorithm that scanned >500,000 radiology and clinical reports for features suggestive of malignant transformation in pancreatic cysts achieved 99.9% sensitivity and 98.8% specificity, effectively maintaining a continuously updated watch list of high-risk patients.¹⁶ This kind of automation reduces the chance that eligible patients are overlooked in busy services.

AI also strengthens radiologic staging. DL models detect subtle imaging patterns beyond human perception.⁴⁵ In prostate cancer, where conventional MRI staging can show limited sensitivity, studies recommend combining advanced MRI sequences with DL analytics to better detect extracapsular extension and other criteria.⁴⁶ In ovarian cancer, a DL pipeline integrating computed tomography/MRI characterized adnexal masses with high sensitivity and specificity, providing risk scores that help MDTs determine surgical need and urgency.⁴⁷ Embedded in radiologists' workflows, these models can prioritize studies, highlight suspicious regions, and standardize measurements, streamlining case preparation and discussion.

Beyond confirmed diagnoses, AI supports early detection and triage.⁴⁸ Prototype systems model symptom-sign constellations using clinical archetypes and terminologies such as SNOMED CT, so that red-flag presentations (e.g., rectal bleeding with weight loss) automatically trigger alerts for expedited colorectal cancer workup and MDT consideration.⁴⁹ Standardized symptom interpretation reduces variability

across clinicians and sites, promoting timely, equitable access to multidisciplinary review.

Across these applications, AI automates front-end tasks, enabling clinicians to focus on interpretation and context, thereby improving operational efficiency, reducing interobserver variability, and preserving essential clinician oversight, judgment, and accountability.

Treatment recommendations and personalized plans

Formulating an optimal treatment plan is the core deliverable of every tumor board (MDT) meeting. Contemporary AI systems emulate MDT reasoning by aligning patient-level data with clinical guidelines and prior cases, thereby generating recommendations that can be compared with—or integrated into—expert judgment.⁵⁰ A prominent example is cognitive clinical decision support such as WFO. After ingesting a patient's clinical profile (diagnosis, stage, biomarkers, comorbidities), WFO outputs ranked options (e.g., “recommended,” “for consideration,” “not recommended”) based on guideline- and literature-derived training.⁵¹ In a multicenter study of colon cancer, WFO's top-tier recommendation exactly matched the board's choice in 48.9% of cases; when the board's choice fell within WFO's “recommended/for consideration” set, concordance rose to 88.4%.⁵² In lung cancer, overall concordance was ~65.8%, reaching 100% in stage IV non-small-cell disease.²⁰ These findings indicate that AI can closely mirror MDT reasoning, particularly where evidence strongly favors specific therapies. Notably, WFO provided no recommendation in ~18% of cases—typically due to idiosyncratic patient factors or underrepresented scenarios—underscoring AI's current role as an adjunctive second opinion, most useful in straightforward cases while remaining challenged by atypical presentations.⁵³

Beyond rule-based systems, LLMs are being evaluated for planning. A custom GPT-4 model for colorectal cancer achieved ~84% agreement with MDT decisions, performing well for standard chemotherapy and surgical choices.⁴² In advanced gastric cancer, an interactive, prompt-optimized LLM improved from ~54.5% to 84% concordance after iterative clarification.⁵⁴ Early studies in head-and-neck cancers similarly show plausible treatment and staging suggestions, though fine-tuning to local standards remains necessary.⁵⁵

AI further enables personalization. Whereas guidelines often follow a one-size-fits-most paradigm, machine-learning models can integrate age, genomics, and patient preferences to signal when deviation is beneficial.⁵⁶ In breast cancer, classifiers predicting the need for chemotherapy versus endocrine therapy achieved AUC 0.94–0.98 in retrospective testing, outperforming some clinical tools and supporting safe treatment de-intensification without compromising outcomes.⁵⁷ AI can likewise stratify patients by expected benefit from targeted agents or trials and, when available, incorporate quality-of-life and preference data (e.g., highlighting fertility preservation for young cervical cancer patients).⁵⁸

Overall, AI functions as a comprehensive, up-to-date knowledge resource that cross-checks MDT deliberations: surfacing relevant trials, emphasizing guideline-concordant options, and reducing the risk of overlooked therapies. Systematic review of divergences between AI and MDT outputs often prompts pro-

ductive discussion and, ultimately, strengthens the robustness of the final treatment plan.

Prognosis prediction and risk assessment

Estimating prognosis is central to MDT deliberations because it informs treatment intensity and follow-up.⁵⁹ AI models, particularly ML and DL, increasingly generate individualized risk estimates that help stratify cases by urgency and aggressiveness of care.^{60,61}

For overall survival and recurrence risk, algorithms trained on large, longitudinal datasets can capture patterns invisible to clinicians. DL/ML prognostic models show solid performance: for example, Diller et al. (2019) predicted survival in adult congenital heart disease with a C-index of 0.735, a proof-of-concept for multifactorial prognostication applicable to oncology.⁶² In hepatocellular carcinoma, an AI-based clinical decision support system that integrated labs, tumor features, and treatments achieved ~82.4% accuracy for 1-year survival, enabling identification of patients at high risk of early mortality for intensified therapy or closer monitoring.⁶³

Beyond survival, AI informs recurrence prediction. In hepatocellular carcinoma, models analyzing preoperative imaging and biopsy data reached a C-index ~0.70 for relapse risk, guiding adjuvant therapy and surveillance.⁶⁴ AI also anticipates treatment response and complications. Ensemble ML in breast cancer flagged cases likely to diverge from guideline-concordant choices. A research reported 43–53% discordance driven by patient-specific nuances (e.g., comorbidities), prompting tailored decisions such as de-escalation of chemotherapy in older adults with limited benefit.⁶⁵ Emerging models predict toxicity (e.g., severe chemotherapy adverse events) from clinical and genetic features, supporting proactive dose modification and supportive care.⁶⁶ Risk modeling can extend to non-cancer-specific hazards (e.g., cardiac events, surgical complications, readmission), triggering earlier specialty involvement and more holistic care plans.⁶⁷

Collectively, AI-derived prognostic and risk tools provide a quantitative backbone for MDTs. High relapse risk may justify escalation; low risk supports de-intensification to preserve quality of life. By precise stratification, MDTs can allocate resources and personalize strategies consistent with the goals of precision oncology.

Cancer-type-specific applications

The impact of AI within MDTs is cancer specific, reflecting distinct decision points, data modalities, and evidence maturity across diseases.⁶⁸ Cancer-type-specific AI modalities, study designs, and primary outcomes are summarized in Table 2.

In lung cancer, AI support is strongest in complex, guideline-driven scenarios. Cognitive decision-support systems have achieved perfect concordance with MDT recommendations in metastatic (stage IV) disease, likely because evidence hierarchies and therapeutic algorithms are clear.⁷⁰ By contrast, concordance drops in early-stage disease and in regions with divergent practice patterns, typically to the mid-60% range, highlighting sensitivity to local standards and incomplete data.^{70,71} LLM approaches add complementary value by triaging pathways—separating curative-intent from palliative trajectories

Table 2. Oncology-specific AI modalities vs. MDT outcomes, by cancer type and study design

Cancer type	AI modality	Study design (as reported)	Primary endpoint	Key result (concordance/AUC/etc.)	Comparative insight	Reference
Colon	Cognitive CDSS (Watson for Oncology)	Retrospective cohort (Korea)	AI-MDT treatment concordance	48.9% strict; 88.4% incl. “for consideration”; N = 656	Broad matching rule inflates concordance vs. strict match	Assessing Concordance With Watson for Oncology, a Cognitive Computing Decision Support System for Colon Cancer Treatment in Korea. ¹⁹
Lung	Cognitive CDSS (Watson for Oncology)	Feasibility, single-center (China)	AI-MDT concordance; unsupported rate	65.8% overall; 100% in Stage IV NSCLC/extensive SCLC; 18.1% unsupported; N = 149 supported	Higher concordance in advanced disease; unsupported cases highlight regional guideline gaps	Using Artificial Intelligence (Watson for Oncology) for Treatment Recommendations Amongst Chinese Patients with Lung Cancer: Feasibility Study. ²⁰
Colorectal	LLM (ChatGPT/ Claude/Llama)	Case-based evaluation	AI-MDT treatment concordance	Up to 84%; N = 95	LLMs approach MDT decisions in typical scenarios; prompt/task design matters	Large Language Models as Decision-Making Tools in Oncology: Comparing Artificial Intelligence Suggestions and Expert Recommendations. ²¹
Gastric	LLM (GPT-4o vs. Gemini Advanced)	Simulation/ iterative prompting	AI-MDT concordance	~54.5% → ~84% after iterative prompting	Human-in-the-loop prompting improves performance	ChatGPT-4o outperforms Gemini Advanced in assisting multidisciplinary decision-making for advanced gastric cancer. ²²
Breast	ML classifiers	Retrospective	Predicting MDT adjuvant drug choices	AUC 0.940 (chemo)/0.899 (endocrine)/0.977 (targeted)	High discriminative ability vs. guideline baselines	Computational prediction of multidisciplinary team decision-making for adjuvant breast cancer drug therapies: a machine learning approach. ²³
Pancreas (risk triage)	NLP (EHR screening)	Large-scale retrospective screen	Detection of high-risk cysts	Sensitivity 99.9%, Specificity 98.8%; 566,233 reports	Automates prescreening/MDT referral; demonstrates scalability	Automated pancreatic cyst screening using natural language processing: a new tool in the early detection of pancreatic cancer. ¹⁶
Liver (HCC)	AI-CDSS/DL	Real-world development/ validation (single health system)	Treatment recommendation & prognosis support	Top-1 accuracy 82.36%; internal MDT match 95.10%	Useful for risk stratification & therapy selection; needs broader external validation	Research of artificial intelligence-based clinical decision support system for primary hepatocellular carcinoma. ²⁴
Multi-condition (Triage)	LLM (GPT-4o/ Claude-3.5/ Gemini-1.5)	Benchmark clinical vignettes	Urgency triage accuracy	92.4% (frontier LLMs; 133/144 cases)	High performance for coarse triage decisions; error bias toward over-triage	The Triage and Diagnostic Accuracy of Frontier Large Language Models: Updated Comparison to Physician Performance. ⁶⁹

AI, artificial intelligence; MDT, multidisciplinary tumor board; CDSS, clinical decision support system; WFO, Watson for Oncology; LLM, large language model; ML, machine learning; NLP, natural language processing; DL, deep learning; NSCLC, non-small-cell lung cancer; SCLC, small-cell lung cancer; AUC, area under the receiver operating characteristic curve; EHR, electronic health record; HCC, hepatocellular carcinoma. Concordance (strict vs. broad): “Strict,” exact match between AI and MDT primary recommendation; “Broad,” MDT’s choice appears among AI’s “recommended/for consideration” options. Study design labels: “Feasibility” denotes single-center pilot/implementation experience; “Case-based evaluation” denotes expert adjudication on curated cases; “Simulation/iterative prompting” denotes offline LLM decision emulation refined through multi-turn prompts.

with >90% accuracy in some studies—thereby channeling cases to the appropriate specialists earlier in the workflow.⁷²

Breast cancer is particularly amenable to AI owing to rich multimodal inputs (genomics, imaging, pathology).⁷³ Supervised machine-learning models reliably predict adjuvant therapy choices, with reported AUCs approaching 0.97, and can standardize risk stratification.⁷⁴ Beyond decision emulation, AI has proven useful on the “output” side of MDTs: systems that draft patient-facing consent and education materials have produced documents consistently rated accurate and comprehensible, potentially improving communication and adherence.⁷⁵

In colorectal cancer, multiple evaluations show high agreement between AI recommendations and MDT decisions.⁴² While strict, top-choice concordance for cognitive systems may hover near 50%, concordance exceeds 85% when acceptable alternatives are included, and independent models—including LLMs—report 85–92% agreement for first-line therapy selection.⁷⁶ This pattern underscores the advantage of AI in cancers with algorithmic care pathways (e.g., criteria for adjuvant chemotherapy in stage II colon cancer), where rapid recall of protocol nuances reduces omissions of standard care.⁷⁷

Evidence in other malignancies is emerging. In gastric cancer, interactive prompting improves LLM concordance from roughly mid-50% to the 80% range, suggesting utility for sequencing multimodal therapy.⁷⁸ For hepatocellular carcinoma, prognostic models (~82% accuracy in some reports) can flag candidates for transplant or intensified therapy.⁷⁹ In prostate cancer, multi-center MRI studies demonstrate high diagnostic accuracy with DL, supporting deployment for imaging review within MDTs.³⁶ Head and neck cancers illustrate the counterpoint: while AI aids staging and recommendation framing, complex anatomic considerations, frailty, and patient preference often tilt decisions toward clinician judgment.⁸⁰

Taken together, cancer-specific analyses converge on a practical rule: the more structured the pathway and the richer the data, the greater the realized AI benefit; the more idiosyncratic the presentation or context, the more AI should be positioned as a safety net and documentation aid—never a substitute for expert consensus.

MDT-AI comparative analysis

Multiple studies have directly contrasted AI-generated recommendations with traditional MDT decisions, providing a meta-level view of AI’s strengths and limitations.⁸¹ Across tumor types and study designs, reported AI-MDT concordance typically ranges from 70% to 90%, indicating that in most cases AI reaches conclusions similar to those of multidisciplinary experts.^{42,82} Discrepancies, when they occur, are often diagnostically informative—exposing gaps in an AI system’s knowledge base or, conversely, practice patterns and local preferences that deviate from the global guidelines on which many systems are trained.^{42,82}

A representative example is WFO in colon cancer, where exact matches with MDT choices occurred in fewer than half of cases.⁸³ Notably, many “discordant” instances reflected MDT selection of options WFO had ranked as secondary or tertiary.⁸³ Such differences often trace to local constraints, clinician experience, or patient-specific considerations that are underweighted in algorithmic logic.⁸³ The converse also arises: AI oc-

casionaly proposes options not initially considered by the MDT.^{40,84} While many are appropriately dismissed after clinical review, some surface emerging therapies or trial-aligned pathways worthy of discussion—functioning as a quality-improvement cross-check that can broaden deliberation.

Beyond concordance, head-to-head comparisons of task accuracy show AI excelling in narrowly defined, controlled tasks—e.g., image classification or pathology subtyping—where single-task accuracies of 90–95% are common.^{85,86} Yet high task accuracy does not automatically translate into clinical utility, because MDT decisions integrate comorbidities, preferences, logistics, and longitudinal context. Importantly, comparative analyses highlight reduced AI flexibility in atypical presentations.⁸⁷ When patient attributes fall outside training distributions (e.g., advanced age), algorithmic recommendations may diverge from geriatric oncology principles; one study attributed ~17% discordance largely to age-related factors.⁸⁸ These edge cases underscore the enduring value of expert judgment.

At the same time, AI can outperform static guidelines—and occasionally expert-only decisions—on consistency and multi-dimensional data integration.^{88,89} For instance, algorithms reliably incorporate genomic results that may be variably weighted in live discussion.⁹⁰ Emerging explainable AI methods further enhance trust by making model rationales legible (e.g., pinpointing features that drove a recommendation), allowing clinicians to calibrate reliance or skepticism appropriately.⁹¹

Overall, direct comparisons highlight a complementary division of labor. AI often replicates expert decisions, excelling in guideline-driven, data-rich contexts, while human expertise is crucial for handling outliers, making trade-offs, and providing context. Hybrid workflows position AI as a supportive, evidence-organizing co-pilot within MDTs, rather than an independent decision-maker.

CHALLENGES AND LIMITATIONS OF AI IN TUMOR MDTs

Technical challenges: accuracy and reliability

AI systems are not uniformly reliable across clinical scenarios. Even with high average performance, small error rates can have serious oncologic consequences. For example, although WFO showed respectable overall concordance for lung cancer, it provided no recommendation in 18.1% of cases, reflecting knowledge gaps or data incompatibility and risking delays if clinicians over-rely on the tool.^{53,92} LLMs can also omit critical safety checks: in one study an LLM failed to flag a major drug-drug interaction (QT-prolongation risk) in 85% of instances when not explicitly prompted.⁹³ These failures—rooted in incomplete training data, prompt sensitivity, or algorithmic oversight—underscore that current AI cannot yet cover the nuanced “corner cases” routinely encountered by MDTs.

Performance frequently degrades when models are deployed outside their development environment due to shifts in patient demographics, scanners, or local practice. A prostate MRI model trained on a single scanner saw AUC fall from 0.95 to 0.83 across external centers and devices.⁹⁴ Likewise, an NLP system for lung cancer pathology that was highly specific (80%) exhibited poor sensitivity (20%) for uncommon subtypes, likely reflecting limited exposure to rare phenotypes during

training.⁹⁵ These domain shifts necessitate site-specific re-validation or retraining—both costly and time-consuming.

Outputs are constrained by input fidelity. Routine clinical data are noisy: EHR fields are missing or inconsistent, and imaging quality varies. The pancreatic-cyst NLP exemplar achieved near-perfect sensitivity partly because it leveraged consistently structured reports; performance would likely erode on poorly formatted notes.⁹⁶ Bias in training corpora is equally consequential. Models trained predominantly on younger or majority populations may underperform in older or underrepresented groups. Notably, WFO's concordance for colon cancer reportedly dropped to ~20% in patients aged >70, indicating inadequate representation of geriatric decision patterns.⁴⁰ Such biases risk entrenching disparities—for instance, under-recommending aggressive therapy to minority patients if historical datasets contain fewer such treatments.⁹⁷ Robust data curation, subgroup auditing, and continuous post-deployment surveillance are therefore essential prerequisites for safe, equitable AI-assisted oncology.

Clinical and ethical issues: data privacy

Many AI systems depend on large datasets and, at times, cloud processing. In oncology, patient privacy is paramount and protected by regulations such as Health Insurance Portability and Accountability Act.⁹⁸ Using web-based AI raises exposure risks; there are documented instances of clinicians inadvertently pasting sensitive data into public tools.⁹⁹ A pragmatic mitigation is local deployment. For example, a locally hosted Llama 3 supporting head-and-neck cancer decisions achieved 92% concordance with the MDT while ensuring data never left secure servers.¹⁰⁰ Institutions must balance AI's benefits against breach risks and favor privacy-preserving designs, acknowledging they are technically demanding and costlier.

When AI influences a clinical decision that ends poorly, responsibility is ambiguous: clinician, hospital, vendor, or algorithm developer? The prevailing view is that AI should function as an adjunct, with ultimate responsibility remaining with human professionals.¹⁰¹ Many centers now require physician verification of AI outputs before action. Legal clarification is advancing: proposals treat AI as a medical device or clinical decision support tool, shifting emphasis to appropriate clinician use. Ethical frameworks such as SEET (safety, efficacy, equity, trust) call for multidisciplinary oversight, transparency about roles, and clear accountability pathways.^{102,103} Until medico-legal standards mature, prudent MDTs will use AI as support—not as an autonomous decision-maker.

Tools must be intuitive; if poorly designed, injecting AI into complex MDT workflows can be disruptive. In rehabilitation settings, staff have resisted AI/robotic systems that changed established routines.¹⁰⁴ In oncology, AI outputs misaligned with board-preferred case formats are easily ignored.¹⁰⁵ Success often hinges on interface customization—embedding AI recommendations into the electronic patient summary under review, or highlighting evidence on presentation slides.^{106,107} Without such seamless integration, AI can slow deliberations. Trust and acceptance are critical: when clinicians cannot see how recommendations are derived, they hesitate. Explainable AI mitigates this by making outputs auditable and discussion-ready while preserving clinical authority.¹⁰⁸

Evidence levels and validation

Despite rapid growth, the evidence base for AI in MDTs still contains gaps that temper confidence and hinder broad adoption.

Most evidence is retrospective, few studies deploy AI prospectively with real-time embedding and outcome tracking.¹⁵ Without randomized or pragmatic trials, causal effects on survival, QoL, time-to-treatment, and costs remain unquantified. For example, WFO has reported colon cancer concordance in retrospective analyses,⁴⁰ but there is no randomized evidence showing that WFO improves outcomes or accelerates care. A small prospective pilot in lung cancer demonstrated feasibility and highlighted failure modes, yet was not powered for endpoints.²⁰ The field needs robust, prospective, ideally multicenter trials to move from plausibility to proof. Such trials are challenging, but they are essential to convert enthusiasm into validated practice.

AI performance is often reported as aggregate metrics, masking variability across clinically important strata.¹⁰⁹ Evidence already suggests sharp performance drops in underrepresented groups—for instance, a marked decrease in WFO concordance among older colon cancer patients.^{110,111} Absent granular subgroup analyses, these limitations remain invisible and unaddressed. Clinicians reasonably expect tools to be tested in the populations they treat, including those with multimorbidity or atypical regimens.¹¹² Without subgroup-specific validation and fairness auditing, any single AI may be applicable only to narrow scenarios, limiting clinical utility and risking inequity.

Process gains do not guarantee patient benefit. An AI can recommend a guideline-concordant therapy that an experienced oncologist would avoid for patient-specific reasons not captured in the model.¹¹³ Longitudinal data on hard endpoints—response rates, toxicity, survival, patient-reported outcomes—are scarce. Until studies link AI-assisted MDT decisions to improved patient outcomes, claims of clinical benefit remain provisional.

AI could plausibly reduce costs by streamlining workflows, minimizing redundant testing, and improving treatment allocation; however, rigorous oncology-specific analyses are lacking.¹¹⁴ A frequently cited example outside oncology showed a 46% cost reduction from an AI hand-hygiene program,¹¹⁵ but analogous evidence in cancer care is minimal. Health systems will require credible cost-effectiveness studies to justify investment in AI for MDTs.

Heterogeneous endpoints, variable definitions of “concordance,” and inconsistent methodological reporting impede comparison and synthesis. The field needs AI-specific reporting standards, analogous to Standards for Reporting Diagnostic Accuracy Studies (STARD) for diagnostics and Consolidated Standards of Reporting Trials (CONSORT) for trials, covering dataset curation, model provenance, performance uncertainty, external validation, calibration, and bias assessment.^{116,117} Recent proposals such as ISLE-ReSt, while not oncology-specific, underscore the value of transparent, reproducible reporting and should inspire fit-for-purpose guidelines for clinical AI.¹¹⁸

Human factors

Introducing AI into clinical practice raises pivotal human-factor questions and risks introducing or amplifying cognitive biases.

Tech-savvy or younger clinicians often view AI as capability-enhancing, whereas others worry that opaque “black-box” models can err in ways that are hard to detect or explain to patients.^{119,120} In one survey of clinicians using an AI decision-support tool, only 17% reported clear improvement in decision-making; many feared over-reliance could erode skills or autonomy.¹¹⁹ Building trust therefore requires more than accuracy gains: models must be transparent; clinicians need education on algorithmic logic and failure modes; and adoption should be grounded in rigorous evidence of safety, generalizability, and clinical utility.¹²¹ Above all, users should experience AI as a tool under their control—not an imposition or surrogate decision-maker.

Automation bias leads users to accept machine suggestions too readily, especially under time pressure. In an MDT, an AI-endorsed plan may be adopted without fully weighing alternatives—the opposite of robust multidisciplinary deliberation.¹²² Algorithmic aversion, triggered by observed errors, can prompt dismissal of otherwise helpful recommendations.¹²³ Mitigation depends on design and process safeguards that keep humans actively engaged: require MDT members to articulate their reasoning even when it aligns with the AI, present confidence estimates and concise rationales to stimulate discussion of uncertainty, and enable overrides with brief justification—an approach shown in medication decision support to preserve human-in-the-loop vigilance.^{113,124} These measures encourage calibrated trust and healthy skepticism.

Competency is equally critical. There is growing momentum to embed AI literacy across medical education.^{125,126} A proposed radiology-residency framework and early “bootcamp” programs demonstrate substantial knowledge gains, with one reporting ~90% post-test accuracy.¹²⁷ Similar, specialty-tailored curricula are needed across oncology. Given MDTs’ collaborative nature, training should be team-based, aligning members on how to interpret outputs, question model assumptions, and incorporate results into shared decisions.¹²⁸

Communication with patients remains underdeveloped. Some may feel uneasy; others may be reassured that advanced tools corroborated clinicians’ plans.^{129,130} Ethical practice favors transparent, plain-language disclosure. With policies still nascent, MDTs should proactively craft strategies that (i) clarify AI’s advisory role, (ii) emphasize clinician oversight and individualized judgment, and (iii) address privacy, limitations, and alternatives. Such framing preserves autonomy while normalizing responsible use.

Adoption in tumor boards demands calibrated trust without over-reliance, bias-aware workflows, and structured team training. Deployment should include clear, patient-centered communication to sustain transparency and shared decisions.

FUTURE DIRECTIONS AND OPPORTUNITIES

Clinical implementation improvements

To translate promising research into routine practice, priorities should shift to prospective, multicenter validation of AI tools and structured implementation programs. Prospective trials are needed to test whether AI-augmented MDTs deliver measurable gains in survival, treatment accuracy, and time-to-treatment.¹³¹

Retrospective analyses indicate, for example, that WFO often mirrors MDT decisions in lung cancer; a forward-looking trial should evaluate whether real-time WFO use accelerates consensus or increases evidence adherence.⁴⁰ Likewise, a multicenter prostate MRI study (AUC 0.91–0.95) demonstrates technical robustness across scanners³⁶; the next step is embedding such imaging AI into live MDT discussions across hospitals to determine effects on decision rates and diagnostic confidence.

AI systems should plug seamlessly into hospital IT, with standardized application programming interfaces (APIs) and EHR compatibility so an oncologist can trigger analyses from the patient chart and view results in the same interface.^{132,133} Institutions should establish real-time feedback loops: when MDTs override AI recommendations, the annotated case is returned to developers to refine models.

Rather than operating in isolation, algorithms should ingest clinician input explicitly.¹³⁴ For instance, an AI could present a ranked differential and allow MDT members to reweight factors (e.g., comorbidities, biomarker uncertainty); the system then updates its recommendations accordingly. Such human-in-the-loop designs combine computational scale with expert judgment in real time.¹¹³ Early studies suggest these collaborative approaches can outperform either AI or clinicians alone in diagnostic tasks (e.g., radiology),¹³⁵ supporting their evaluation in oncology MDTs.

Education and training

Curricula are being developed to ensure AI literacy among new professionals. Oncology fellowships will increasingly add modules on interpreting model outputs, fundamentals of algorithm development, and ethical-legal considerations.¹³⁶ Interdisciplinary workshops linking oncologists, data scientists, and ethicists can build shared understanding and enable co-design of clinician-centric tools.¹³⁷ Professional societies may introduce certifications or continuing medical education (CME) focused on oncology AI, standardizing competence. Notably, a pilot radiology design-sprint curriculum increased residents’ AI knowledge accuracy to >90%¹³⁸; scaling such programs will be essential for broad, confident adoption.

Ethical and policy aspects

Globally, regulators are converging on governance frameworks, yet their approaches differ in ways shaping oncology MDTs implementation.

Adopted in 2024, the EU AI Act is the first comprehensive cross-sector law regulating AI.¹³⁹ It applies a risk-based taxonomy that classifies most medical AI systems as high risk, subjecting them to stringent obligations for risk management, data quality, robustness, transparency, human oversight, and continuous post-market monitoring.¹³⁹ For oncology, this means any AI tool used in MDT deliberations (e.g., diagnosis support or treatment planning software) must provide extensive technical documentation, clear intended-use statements, and undergo conformity assessment before deployment.¹³⁹ The Act’s extra-territorial reach further extends these duties to providers outside the EU whose tools are used within the bloc.^{139,140} In practice, compliance pressures are likely to drive developers toward more transparent, auditable, and resilient systems; for MDTs,

this should translate into certified tools labeled with explicit indications, performance bounds, and limitations, akin to regulated medical devices.

The US Food and Drug Administration has proposed a Total Product Life Cycle (TPLC) approach for AI/ML-based medical devices that emphasizes ongoing oversight rather than one-time clearance.¹⁴¹ Manufacturers are encouraged to adopt Good Machine Learning Practice and, where appropriate, submit Predetermined Change Control Plans detailing how algorithms may adapt safely post-authorization.^{141,142} The agency is also expanding reliance on real-world evidence and post-market performance monitoring.^{141,143} For US oncology MDTs, this implies AI tools may evolve iteratively under controlled updates, with obligations to collect and analyze data on recommendations versus clinical outcomes.

Complementing statutory regimes, World Health Organization's 2021 guidance articulates six principles for AI in health that foreground human rights and public interest.¹⁴⁴ For MDTs, these principles underscore augmentation over replacement: AI should make reasoning traceable and be trained and validated on diverse cohorts to avoid performance gaps across patient subgroups. The guidance also stresses community engagement and respect for patients' rights, suggesting future assessments should weigh acceptability alongside technical merit.¹⁴⁴

Taken together, these regulatory and ethical trajectories point toward oncology AI that is safer, more standardized, and more patient-centered. In practical terms, MDTs may increasingly rely on certified decision-support tools with clear use cases, explicit performance disclosures, monitored updates, and built-in explainability.

Potential impacts on healthcare systems

If AI integration into MDTs proceeds successfully, the consequences for healthcare could be substantial across outcomes, efficiency, equity, and sustainability.

Better decision support should translate into better patient outcomes. By helping MDTs detect errors or surface overlooked options, AI could raise survival and lower recurrence.^{145,146} For example, an enhanced prognostic model—such as a DL system with a C-index near 0.75—may identify patients who truly require additional therapy, thereby reducing mortality by guiding intensified treatment to those most likely to benefit.¹⁴⁶ Although the precise effects on survival and quality of life remain to be proven prospectively, that remains the ultimate objective.

AI can make care more efficient by automating data gathering and first-pass analysis, shortening per-case preparation and enabling discussion of more cases within the same time window.¹⁴⁷ Cost-effectiveness may improve if AI helps avoid futile or redundant interventions—for instance, flagging patients unlikely to benefit from expensive drugs or preventing unnecessary tests. Evidence from outside oncology shows cost reductions (e.g., automated hand-hygiene monitoring; AI-assisted diabetes management that fine-tunes therapy and improves outcomes).^{115,148} In oncology, analogous benefits could include optimizing follow-up intervals or recommending the most cost-effective therapy among similarly effective options. These gains, however, must be balanced against acquisition and maintenance costs and the need for robust IT infrastructure.

AI could extend specialty expertise to regions lacking MDTs. A cloud-based system might support community oncologists with input approximating an academic tumor board, aligning decisions with best evidence.¹⁴⁹ Conversely, if advanced tools remain concentrated in wealthy centers or underperform in underrepresented populations, disparities may widen.¹⁵⁰ Ensuring equity will be essential—potentially via open-source or low-cost solutions and systematic fairness evaluations.¹⁵¹

AI may also advance environmental sustainability. More efficient workflows reduce waste; “green oncology” initiatives could leverage AI to optimize scheduling, consolidate visits, and minimize unnecessary travel (e.g., triaging which patients require central tumor board review versus local management with AI guidance).^{152,153} These applications remain speculative but are increasingly relevant as health systems assess their carbon footprints.¹⁵³

Ultimately, progress will depend on sustained collaboration among clinicians, researchers, policymakers, and patients. By prioritizing rigorous validation, clinician education, ethical governance, and health-system evaluation, AI can evolve into an integral, trusted partner in cancer care. These trajectories point toward safer, more standardized, and more patient-centered oncology AI systems (Figure 3).

Limitations of the study

This work is a narrative synthesis focused on how AI can support MDTs, rather than a registered systematic review or meta-analysis. As such, study selection may be subject to publication and language biases, and effect sizes across reports are not pooled. The underlying evidence base remains heterogeneous: many studies are single-center, retrospective, or simulation-based; endpoints vary widely (e.g., task accuracy, guideline concordance, time savings), and few prospective multicenter evaluations link AI-assisted MDT decisions to patient-centered outcomes such as survival, toxicity, or quality of life. Consequently, cross-study comparisons should be interpreted cautiously.

Generalizability is a key constraint. Datasets often over-represent high-income settings, common cancers, and specific imaging or EHR infrastructures, raising risks of spectrum and domain shifts when models are deployed across hospitals, scanners, or populations. Reporting on fairness is inconsistent; impacts across sex, gender, age, ethnicity, and socioeconomic groups are rarely quantified, and calibration under distribution shift is infrequently assessed.

Model transparency and reproducibility also limit adoption. Proprietary systems, incomplete reporting of preprocessing and hyperparameters, and restricted access to code, weights, and data impede independent verification. LLMs are sensitive to prompts and context windows and can generate confident but incorrect outputs; systematic error analyses, human-factors testing, and robust guardrails are not yet standard.

Finally, real-world implementation evidence remains sparse. Workflow integration, accountability, automation bias, alert fatigue, and clinician training are incompletely studied, as are legal, privacy, and cross-border data-governance constraints. Economic evaluations and maintenance plans for post-deployment monitoring are limited. Our regulatory framing emphasizes



Figure 3. Future directions and opportunities for AI-integrated tumor boards

The infographic illustrates the bidirectional collaboration between AI systems and multidisciplinary clinical teams (MDTs) through a continuous data feedback loop. Four thematic domains are outlined: (1) clinical implementation improvements (interoperability, multicenter trials, EHR-based AI modules), (2) education and training (AI literacy, interdisciplinary collaboration), (3) ethical and policy aspects (EU AI Act, FDA TPLC, WHO guidelines), and (4) healthcare system impact (survival, equity, cost efficiency, sustainability). The circular layout and holographic flow elements symbolize real-time information exchange and adaptive learning between AI and clinical practice.

widely cited approaches (e.g., life cycle oversight), which may not capture all regional pathways.

Together, these limitations temper the strength of causal claims about clinical benefit. We therefore prioritize a research agenda of prospective multicenter trials, standardized reporting and fairness audits, external validation under temporal and geographic shifts, cost-effectiveness analyses, and continuous post-market performance and safety monitoring within clear governance.

CONCLUSION

AI has become a powerful adjunct to MDTs, enhancing decision-making and streamlining care. Across ML, NLP, DL, and LLMs, AI frequently aligns with expert recommendations, with overall concordance often 70–90%. Decision-support systems report ~65–88% agreement in lung and colon cancer, and up to ~84% in simulated colorectal and gastric decisions. AI also automates diagnostics and staging (e.g., 72–94% accuracy for lung cancer TNM from pathology) and personalizes therapy choices. By standardizing inputs and offering a consistent evidence-based second opinion, AI reduces interobserver variability and optimizes MDT workflow—e.g., NLP can synthesize thousands of records within seconds, enabling earlier detection (near-100% sensitivity for pancreatic lesion screening) and more complete case reviews. Benefits extend to patient communica-

tion (fully actionable consent drafts) and predictive accuracy (ML AUC 0.94–0.98 for breast adjuvant decisions).

DATA AND CODE AVAILABILITY

This narrative review did not generate or analyze any new datasets and did not develop or train any custom software or models. All evidence summarized in the text, figures, and tables (Figures 1, 2, and 3; Tables 1 and 2) was extracted from publicly available, peer-reviewed studies indexed in PubMed and Web of Science. No proprietary or non-public datasets were accessed. Because this work synthesizes published findings, data availability is subject only to the sharing terms of the original sources.

No custom code was used beyond standard office/graphics software for figure preparation and manuscript formatting. Study-level extraction worksheets (e.g., items compiled to produce Tables 1 and 2) and the bibliographic search strings can be provided by the corresponding author on reasonable request.

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AUTHOR CONTRIBUTIONS

All authors contributed to the conception, design, and execution of this review manuscript. X.W. contributed to the literature search, data curation, and drafting of sections on AI applications in diagnostics and staging (diagnostic and staging support) and cancer-type-specific applications (cancer-type-specific applications). Q.W. was involved in formal analysis of evidence from retrospective and prospective studies, visualization of tables (Tables 1 and 2), and writing the sections on treatment recommendations (treatment recommendations and personalized plans) and prognosis prediction (prognosis prediction and risk assessment). G.D. participated in methodology development, validation of AI performance metrics, and revising the overview of AI technologies (overview of AI technologies in oncology) and challenges (challenges and limitations of AI in tumor MDTs). J.W. contributed to the investigation of ethical and policy aspects and drafting the future directions (future directions and opportunities) and conclusion (conclusion). Y.T. assisted with resources, software tools for data synthesis, and editing the introduction (introduction) and abstract. Y.F., as the corresponding author, supervised the project, conceptualized the overall structure, performed project administration, and finalized the manuscript with critical revisions for intellectual content. All authors reviewed and approved the final version of the manuscript.

DECLARATION OF INTERESTS

The authors do not have any conflicts of interest to disclose.

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